Two Margins of International Athlete Recruitment to NCAA Division I: Entry, Sublinear Scaling, and Shock-Response Case Studies

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Abstract

We construct an international NCAA Division I athlete panel from a 2024–25 cross-section of 19,317 track-and-field and soccer rosters, an eight-sport multi-year panel of 260,737 athlete-years (2019–2023), and a subsample of TFRRS individual season-best marks. **On the extensive margin**, log GDP per capita and log population predict whether a country sends any athletes ($\hat{\beta} = 0.52$, 0.18; both $p < 0.01$); political stability does not. **On the intensive margin**, athletes scale sublinearly with source-country population: cohort elasticities of 0.33 (OLS), 0.48 (NegBin), and 0.61 (Heckman, selection-corrected). The political-stability coefficient is fragile across specifications and, in the eight-sport panel, the cross-sectional +0.34 ($p < 0.01$) collapses to -0.005 once country fixed effects are added—*the cross-country result is entirely between-country composition*. A COVID natural experiment from differential sport disruption yields a -0.143 ($p < 0.01$) transitory effect for soccer and volleyball relative to other sports. **Four shock case studies** examine named events. Russian (N=195, $p < 0.001$) and Ukrainian (N=83, $p < 0.01$) under-classman athletes are 16 and 8 percentage points *more* likely to return to their U.S. school after the 2022 invasions, relative to a same-region donor pool—consistent with a destination-substitution channel where home exit options vanish. The 2020 Lebanese financial collapse instead cuts new arrivals by $\sim$ 70% ($\log \beta = -1.22$, $p < 0.01$) without a detectable change in retention—an extensive-margin channel operating through household funding capacity. Chronic Venezuelan instability shows neither a sharp inflow change nor a retention boost but a slow retention erosion (0.60 to 0.41 over four years).

1 Data

We work with two complementary roster datasets. The **cross-section** (used in Sections 4.1–4.5 and 4.8) is a country-level panel of international student-athletes currently enrolled in NCAA Division I rosters in the 2024–25 / 2025–26 academic season. Athlete-level records are scraped from official athletic-department websites for soccer (660 athletes from 27 schools) and track and field / cross country (18,657 athletes from 328 of 355 D-I programs; school coverage 92.4%). Only 68 of 1,830 international athletes compete in soccer, so we report the combined sample in headline tables but the empirical setting in Sections 4.1–4.5 is the international track-and-field market specifically.

The **2019–2023 multi-sport panel** (introduced in Section 4.7, used in Sections 4.6, 4.7, and 4.9) extends the roster scrape to eight sports (track, soccer, basketball, swimming, tennis, golf, ice hockey, volleyball) across 2019–2023 D-I athletic-department snapshots, yielding 260,737 athlete-years. A separate cross-sectional pull adds NCAA Academic Progress Rate (121K records, 2005–2025) and Graduation Success Rate (117K records, cohorts 1998–2018) and an IPEDS + EADA join to institutional Carnegie classification, nonresident-alien enrollment share, and athletic finances. Individual TFRRS season-best marks add an athlete-event-year performance subsample for 244 international track athletes.

Hometown text is parsed from each roster page and mapped to ISO-3 country codes via a normalisation procedure that combines US state abbreviations (both postal and AP-style), Canadian provinces, World Bank country names, and a manual alias map for common roster forms. The mapping resolves 87.6% of track and 99.8% of soccer hometowns.

Country covariates come from World Bank WDI (GDP per capita PPP, total population, 15–24-year-old cohort 2018–2022 average, CPI inflation, merchandise terms of trade), Worldwide Governance Indicators (political stability, government effectiveness, rule of law, control of corruption), UCDP/PRIO conflict, EM-DAT natural disasters, the Oxford COVID-19 Government Response Tracker, and UNHCR Refugee Statistics. We use 2023 cross-sections and 2010–2023 cumulative aggregates. The intensive-margin analysis covers the **102 countries with at least one international athlete**; the extensive-margin probit uses a wider universe of 191 countries with non-missing covariates. We assign each country to one of eight regions for the regional fixed-effects analysis in Section 4.8: Sub-Saharan Africa, MENA, South Asia, East Asia & Pacific, Latin America & Caribbean, North America, Western Europe, and Eastern Europe & Central Asia.

Table 1. Descriptive statistics by income tier

Countries split into quartiles by 2023 GDP per capita PPP.

Tier (Q of GDP/cap PPP)	Countries	Athletes	Per country	GDP/cap PPP (med.)	Athletes/mil 15–24	Pol. stal
Q1 Low	26	533	20.5	\$7,670	0.84	
Q2 Lower-middle	25	109	4.4	\$21,917	2.38	
Q3 Upper-middle	25	352	14.1	\$41,288	12.15	
Q4 High	26	825	31.7	\$64,485	13.17	
All	102	1819	17.8	\$30,837	5.59	

2 Background

The paper sits at the intersection of four literatures: the economics of migration selection, the economics of international student mobility, the economics of college athletics labor markets, and the empirical literature on how migrants respond to home-country shocks.

Migration and self-selection. The Roy model framework introduced to migration by Borjas (1987) predicts that who moves and from where is governed by relative returns to skill across origin and destination. The empirical test in Grogger and Hanson (2011) on aggregate bilateral flows shows positive selection of educated migrants from poor countries, with the educated cohort drawn disproportionately from countries with higher per-capita GDP—a sublinear scaling of migration in source-country population that mirrors what we report for NCAA international recruitment.

The classical Roy logic is about *whether* migrants come; the sublinear-scaling contribution here is about *how many* come once a country is in the pipeline. Borjas and Bratsberg (1996) studies the complementary margin of return migration, an outcome which our case studies in Section 4.9 touch but cannot cleanly estimate.

International student mobility. Bound et al. (2015) examine the response of foreign STEM doctoral applications to U.S. labor-market conditions and OPT policy, documenting that international students are sensitive to post-degree opportunity rather than only to direct cost. Bound et al. (2021) extend this to the bachelor's margin and the response to immigration-policy uncertainty in 2016–2020. The NCAA international recruitment market is a specialised slice of this broader student-mobility literature with two distinguishing features: (i) athletic scholarships provide a separate funding channel that breaks the financial-constraint mechanism emphasised in much of the international-student work, and (ii) recruitment intensity is position- and event-specific, producing pipelines that the country-level migration literature does not naturally accommodate.

Sports labor markets and college athletics. Sanderson and Siegfried (2018) survey the economics of college athletics, with particular emphasis on labor-market questions that the NCAA's amateurism regime makes salient. Empirical work on international labor in U.S. professional leagues (Berri and Schmidt, 2007) studies productivity of international NBA players; the equivalent at the college level has been largely descriptive. Brown (1993) provides the foundational empirical work on the determinants of athletic recruitment at the program level. Our position $\times$ institution sorting result in Section 4.6 maps to this literature: deliberate pipelines (Nigerian basketball centers, Australian football K/P) sort to specific high-resource programs in a pattern invisible at the aggregate sport level.

Networks, pipelines, and shock response. Munshi (2003) establishes that established migrant networks materially affect new entrants' outcomes in the host labor market; Patel and Vella (2013) extends the network logic to occupational sorting. The country-event specialisation we document in Section 4.5 (Kenya–Iten in distance, Jamaica–MVP in sprints, the Netherlands multi-event tradition) is a network manifestation of the same logic in a sports-specific market. On shock response, Yang (2008) uses Philippine exchange-rate shocks to show that household-level economic shocks change remittance and investment behaviour; Bohra-Mishra and Massey (2017) documents how active armed conflict affects educational attainment of children left behind. The case studies in Section 4.9—Russia 2022, Ukraine 2022, Lebanon 2020—are direct analogues at the college-athlete margin: war and sanctions raise retention through closed exit options, while financial collapse cuts new arrivals through funding capacity.

Contribution. The paper's three main contributions are (i) a clean two-margin decomposition of international athlete supply with sublinear cohort population elasticity (0.33–0.61) and a documented composition-vs-within-country distinction for governance; (ii) a within-event analysis that shows the country-pool average masks economically distinct supply structures across event groups; and (iii) four named natural experiments that identify direction- and margin-specific shock-response patterns where a population-level panel specification does not.

3 Specification

We decompose international NCAA athlete supply into two margins. The extensive margin is whether country c sends any athletes,

$$\Pr(\text{any athletes}_c = 1) = \Phi(\alpha + \gamma_1 \log \text{GDPpc}_c + \gamma_2 \log \text{pop}_c + \gamma_3 \text{polstab}_c)\,. \tag{1}$$

The intensive margin is the count of athletes among senders. We use $\log \text{pop}_{15\text{–}24,c}$ as a free regressor rather than a -1 offset:

$$\log \text{athletes}_c = \alpha + \beta_1 \log \text{pop}_{15\text{–}24,c} + \beta_2 \log \text{GDPpc}_c + \beta_3 \text{polstab}_c + \mathbf{x}_c' \boldsymbol{\delta} + \varepsilon_c. \tag{2}$$

Section 4.3 documents why a rate outcome (athletes/cohort) with log pop omitted gives misleading results. We complement OLS with (i) a Heckman two-step selection correction using the probit from Equation 1, (ii) a Negative Binomial specification on raw counts, which handles low-count truncation that OLS on log athletes does not, and (iii) quantile regression at $\tau \in \{0.25, 0.50, 0.75\}$. We report Huber–White (HC1) robust standard errors for OLS.

4 Results

4.1 Extensive margin

	Probit: Pr(sends any athletes)
log GDP per capita PPP	$+\mathbf{0.52}^{***}$ (0.11)
log population	$+\mathbf{0.18}^{***}$ (0.06)
Political stability	+0.19 (0.17)
n	191
Pseudo R^2	0.19

GDP per capita and population determine entry; political stability does not. Marginal effect of log GDPpc on Pr(sends) is approximately +0.20 per log-unit at the sample mean.

4.2 Intensive margin: three specifications

Different estimators give different answers on the intensive margin. We report all three side by side.

Coefficient	OLS log-count	Negative Binomial	Heckman 2-step (w/ IMR)
$\log \text{pop}_{15\text{–}24}$	$+\mathbf{0.33}^{***}$ (0.11)	$+\mathbf{0.48}^{***}$ (0.15)	$+\mathbf{0.61}^{**}$ (0.29)
log GDPpc	$+\mathbf{0.37}^{**}$ (0.16)	−0.07 (0.18)	+1.44 (0.93)
Political stability	+0.37 (0.25)	$+\mathbf{0.79}^{***}$ (0.25)	$+0.60^{*}$ (0.34)
$\log(1 + \text{disaster events})$	+0.03 (0.19)	−0.01 (0.26)	+0.10 (0.19)
Inverse Mills ratio	—	—	+3.27 (2.64)
n	100	100	100
R^2 / pseudo-R^2	0.21	—	0.22

The sublinear-scaling fact is robust across estimators. The cohort population elasticity ranges from 0.33 (OLS) to 0.61 (Heckman). All three estimates are well below the unit elasticity that a constant-returns supply curve would predict. The Heckman estimate, which corrects for selection into the sender pool, sits in the middle of the reviewer-flagged range (0.5–0.7) and is our preferred point estimate. The substantive conclusion is the same across methods: doubling the source-country 15–24 cohort raises NCAA athlete supply by considerably less than a factor of two.

Political stability is not pinned down. The OLS coefficient (+0.37, SE 0.25) is statistically zero, but the data have only $\sim 80\%$ power to reject effects larger than MDE $= 1.96{\cdot}0.25{+}0.84{\cdot}0.25 \approx 0.70$ (Cohen, 1988). The NegBin estimate is +0.79 ($p < 0.01$); the Heckman estimate is +0.60 ($p < 0.10$). Both are near the OLS MDE. The most defensible reading is that we cannot distinguish between zero and a moderately positive effect of about +0.7; the OLS-zero, NB-positive, and Heckman-marginal estimates are all within sampling variability of one another.

Log GDP per capita is fragile across estimators. OLS gives +0.37 ($p < 0.05$); NegBin gives −0.07 (n.s.); Heckman gives +1.44 with SE 0.93 (n.s., but the point estimate is large). Wealth is unambiguously important on the extensive margin (Section 4.1); on the intensive margin its role is estimation-method-dependent and we do not claim a precise effect.

Disaster events do not survive any specification. The coefficient is essentially zero in OLS, NegBin, and Heckman. The previous draft's -0.74^{***} in the rate model was an artefact (Section 4.3).

4.3 Methodological note: the rate-model artefact

A previous version of this paper used $\log(\text{athletes}/\text{pop}_{15-24})$ as the intensive-margin outcome and *omitted* $\log \text{pop}$ from the right-hand side. That specification implicitly imposes a unit elasticity of athletes wrt population. The data sharply reject this restriction: the free-elasticity estimate is between 0.33 and 0.61 depending on the estimator, not 1.0. The mis-specification gets loaded onto the other coefficients.

Coefficient	Rate model (implicit $\beta_{\text{pop}} = 1$)	Count model (β_{pop} free, OLS)	Difference
$\log \text{GDPpc}$	+0.25 (0.22)	$+0.37^{**}$ (0.16)	—
Political stability	$+\mathbf{1.12}^{***}$ (0.27)	+0.37 (0.25)	**collapses (OLS)**
$\log(1 + \text{disaster events})$	$\mathbf{-0.74}^{***}$ (0.16)	+0.03 (0.19)	**collapses**
$\log \text{pop}_{15-24}$	(constrained = −1)	$+\mathbf{0.33}^{***}$ (0.11)	—
R^2	0.53	0.21	—

Small countries (Norway, Iceland, Switzerland, Bahamas, Jamaica) are systematically high on the WGI political-stability scale and have relatively few EM-DAT disaster events; large countries (India, China, Indonesia, Pakistan) have the opposite. When the rate model constrains the population elasticity to −1, this small-vs-large correlation loads onto the stability and disaster coefficients. The collapse from +1.12 to +0.37 in OLS—and to +0.79 in NegBin or +0.60 in Heckman—is what the artefact correction looks like.

We retract the previous draft's headline that "political stability dominates the intensive margin" and replace it with the more cautious reading in Section 4.2: stability may matter, but our identification of its magnitude is loose.

4.4 Quantile regression: heterogeneity in the population elasticity

OLS estimates the conditional mean; if the country-pipeline residual story is right, the population-supply relationship may differ across the distribution of athlete supply.

Coefficient	$\tau = 0.25$	$\tau = 0.50$	$\tau = 0.75$
$\log \mathrm{pop}_{15\text{–}24}$	$+0.24^{**}$	$+\mathbf{0.42}^{***}$	$+\mathbf{0.51}^{***}$
log GDPpc	+0.28	$+0.49^{*}$	+0.39
Political stability	+0.21	+0.46	$+0.78^{*}$

The population elasticity **more than doubles** from $\tau = 0.25$ to $\tau = 0.75$ ($0.24 \to 0.51$). Among countries supplying few international athletes per cohort (the lower quartile), the relationship is very weak; among high-supply countries (the upper quartile), the relationship is closer to proportional. The OLS average of 0.33 smooths these together and probably understates both the floor and the ceiling. Political stability also moves strongly across the distribution (+0.21 at $\tau = 0.25$ to +0.78 at $\tau = 0.75$), suggesting institutional quality matters mostly for countries already supplying many athletes, where the marginal unit of governance can extend the existing pipeline rather than build one from scratch.

4.5 Sport-event heterogeneity

The international concentration is in track. Our roster data include a `position` field that we parse into nine standard event groups (Distance, Sprints, Throws, Jumps, Mid-Distance, Hurdles, Multi-Events, Pole Vault, and Cross Country) for the 1089 international track athletes whose schools surface event information. This section uses the within-event variation to ask whether the country-panel pooled result of Section 4.2 is masking event-specific structure.

Event composition and pipeline concentration

Event group	n	Share	Top-5 source countries (share)	Region HHI
Distance	409	37.6%	KEN 25 %, GBR 15 %, CAN 11 %, AUS 8 %, ESP 5 %	0.253
Sprints	211	19.4%	NGA 18 %, JAM 15 %, CAN 7 %, BHS 6 %, GBR 5 %	0.251
Throws	136	12.5%	JAM 16 %, CAN 9 %, GBR 7 %, ZAF 6 %, NZL 5 %	0.218
Jumps	120	11.0%	NGA 9 %, CAN 8 %, JAM 7 %, FRA 5 %, GBR 5 %	0.177
Mid-Distance	73	6.7%	GBR 18 %, CAN 18 %, NZL 8 %, AUS 8 %, ESP 8 %	0.252
Hurdles	61	5.6%	JAM 15 %, GBR 7 %, DEU 7 %, BHS 7 %, BRB 5 %	0.217
Multi-Events	55	5.1%	GBR 16 %, NLD 13 %, CZE 7 %, IRL 7 %, ESP 5 %	0.559
Pole Vault	24	2.2%	RUS, GRC, ITA, SWE, CAN (each 8 %)	0.302

Two facts stand out. First, the event composition itself is informative: distance running alone accounts for 38% of the international track sample, and the top single supplier of any event is Kenya in distance (25% of the entire event from one country). Second, regional Herfindahl indices (computed over our eight-region classification) show that supply is moderately concentrated for most events (HHI $\approx$ 0.18–0.30) but extremely concentrated for Multi-Events (HHI = 0.56, with Western Europe supplying 73% of all international multi-event athletes). The Caribbean sprint and East-African distance networks produce the second (HHI = 0.25) and third-highest concentrations after multi- events, though they share their event group with other meaningful regional clusters (Sub-Saharan Africa for sprints, Western Europe for distance).

Regional shares by event

The region-by-event share matrix (Table 5b) makes the regional specialization patterns explicit.

Region	Distance	Hurdles	Jumps	MidDist.	Multi	PV	Sprints	Throws
Sub-Saharan Africa	**26.7**	18.0	**22.5**	4.1	0.0	4.2	**30.8**	12.5
Latin Am. & Caribbean	2.4	**32.8**	13.3	4.1	1.8	8.3	**33.6**	**27.9**
Western Europe	**38.9**	24.6	**27.5**	**42.5**	**72.7**	**50.0**	19.0	**32.4**
Eastern Europe & C.A.	4.6	1.6	13.3	8.2	16.4	16.7	2.8	8.1
East Asia & Pacific	12.2	8.2	5.8	16.4	3.6	8.3	2.4	5.9
MENA	3.9	8.2	6.7	5.5	1.8	4.2	2.4	3.7
North America	11.0	4.9	8.3	17.8	3.6	8.3	7.1	8.8
South Asia	0.0	1.6	2.5	1.4	0.0	0.0	0.9	0.0

Sub-Saharan Africa supplies 27% of distance running and 31% of sprints—the classical pipelines (Kenya, Ethiopia for distance; Nigeria for sprints and jumps). **Latin America & Caribbean** contributes one-third of sprints (34%) and one-third of hurdles (33%), the Jamaica/Bahamas/Trinidad/Barbados block. **Western Europe** dominates the technical events: multi-events (73%), pole vault (50%), mid-distance (43%), and supplies nearly 40% of distance overall (the European distance scene plus a long tail of contributions to every event). South Asia is essentially absent from every event, despite very large populations—more on this in Section 5.

Within-event count regressions

Table 6 reports the within-event count specification (Equation 2) for all eight event groups with enough sender countries to fit, using both OLS on log(athletes) and Negative Binomial on raw athlete counts. The NB column is the preferred reading because it handles low-count truncation that OLS on log(athletes) does not.

	OLS log-count				Negative Binomial			
Event	$\log \text{pop}_{15-24}$	$\log$ GDPpc	polstab	R^2	$\log \text{pop}_{15-24}$	$\log$ GDPpc	polstab	n
Distance	$+0.39^{***}$	-0.46	$+0.93^{**}$	0.23	$\mathbf{+0.64}^{***}$	$\mathbf{-0.74}^{**}$	$\mathbf{+1.13}^{***}$	40
Sprints	$+0.13^{**}$	$+0.05$	$+0.13$	0.07	$+0.13$	-0.21	$+0.11$	43
Throws	$+0.18^{**}$	-0.11	$+0.30$	0.10	$\mathbf{+0.25}^{**}$	$\mathbf{-0.57}^{**}$	$\mathbf{+0.69}^{**}$	45
Jumps	$+0.10$	-0.05	$+0.06$	0.06	$+0.11$	-0.02	$+0.02$	42
Mid-Distance	$+0.30^{**}$	$+0.10$	$+0.57^{**}$	0.33	$\mathbf{+0.45}^{***}$	$+0.07$	$\mathbf{+0.92}^{**}$	24
Hurdles	-0.06	-0.09	$+0.28$	0.17	-0.06	-0.24	$+0.43$	31
Multi-Events	$+0.19^{*}$	$+0.45^{**}$	$+0.38$	0.33	$+0.26^{*}$	$+0.55$	$+0.44$	22
Pole Vault	-0.03	$+0.34^{**}$	-0.26^{*}	0.13	-0.03	$+0.37$	-0.28	19

*HC1 SEs (OLS); MLE SEs (NB). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.*

Three distinct event types emerge.

(A) Low-GDP, high-stability events: Distance, Throws. In both NegBin specifications, the log GDP per capita coefficient is *negative* and significant (-0.74^{**} for distance, -0.57^{**} for throws), and political stability is large and positive ($+1.13^{***}$ and $+0.69^{**}$). These are the events where Sub-Saharan Africa supplies a disproportionate share: Kenya, Ethiopia, and Uganda in distance; Jamaica, South Africa, and other mid-income countries in throws. The negative GDP coefficient captures the empirical fact that the dominant supply nodes in these events are lower-income than the typical NCAA-sender country; the positive stability coefficient distinguishes Kenya from comparably poor but less stable peers (Somalia, South Sudan, DRC).

(B) High-GDP-or-mixed events with stability premium: Mid-Distance, Multi-Events. Mid-distance running shows a textbook Anglo-sphere pattern: positive population elasticity ($+0.45^{***}$), zero GDP coefficient, and a large positive stability coefficient ($+0.92^{**}$). The supply is dominated by GBR, Canada, New Zealand, Australia, and Spain. Multi-events show the cleanest positive GDP gradient ($+0.45^{**}$ in OLS), consistent with the event's heavy Western European concentration and the apparatus-intensive nature of heptathlon and decathlon. Pole vault similarly favours higher-GDP countries ($+0.34^{**}$).

(C) Caribbean-clustered events with no macro signal: Sprints, Hurdles, Jumps. For each of these events, neither GDP nor political stability is distinguishable from zero in either specification, and the population elasticity is small or zero. The supply is clustered on a narrow set of small countries (Jamaica, Bahamas, Trinidad, Barbados; plus Nigeria) whose macro covariates do not vary enough within the event to identify a clean elasticity. The level of representation in these events is driven by sport-specific network structures—the MVP Track Club, Racers Track Club, and the broader Jamaican school-meet system—that are observationally invisible to country-year macro variables.

Country-event specialisation

We can also examine which countries supply more athletes in each event than their macro covariates would predict. Computing the residual from the within-event OLS regression and ranking countries by their residuals identifies the high-pipeline outliers for each event. The top three positive residuals are:

Event	Top three positive residuals
Distance	Kenya, Ethiopia, Uganda
Sprints	Jamaica, Bahamas, Nigeria
Throws	Jamaica, New Zealand, South Africa
Mid-Distance	Great Britain, Norway, New Zealand
Multi-Events	Netherlands, Great Britain, Czech Republic

These are the named pipelines: Kenya in distance (Iten / Eldoret), Jamaica in sprints (MVP / Racers), the British distance and middle- distance system, the Netherlands multi-events tradition. Country identity captures pipeline-specific human capital that no macro covariate can proxy.

What the event split tells us

The country-panel pooled results of Section 4.2 average over three economically distinct supply structures: a low- GDP / high-stability East-African distance and throws complex; a high-GDP-or-stable Anglosphere mid-distance and technical-event complex; and a Caribbean sprint network whose supply is invisible to macro variables. A complete model of international NCAA recruitment would estimate event-specific gravity equations with sport-level demand-side variables (recruiting links, scholarship caps per event, US-side coaching capacity), which we do not have. The practical implication for our country-level results is that the sublinear population elasticity of ~ 0.4 is a weighted average of mostly-East-African and mostly-European structures that may have different true elasticities individually.

4.6 Position structure and institutional sorting across team sports

Extending the within-event approach to team sports, we classify position fields for seven additional sports using regular-expression rules on the free-text `position` variable in the eight-sport roster panel: soccer (GK/D/M/F), basketball (G/F/C plus G/F and F/C hybrids), football (QB/RB/WR/TE/OL/DL/LB/DB/ST), volleyball (OH/MB/OPP/S/L-DS), baseball/softball (P/C/INF/OF/UT and ice hockey (G/D/F). Two facts stand out.

Position-specific country pipelines. Several positions draw disproportionately from one source country, in patterns invisible at the sport level.

Sport	Position	% intl	Top sources	Top region (share)
Basketball	Center (C)	23.7%	NGA 33, CAN 22, DEU 13	SSA (29%)
Football	Kicker/punter (ST)	5.1%	AUS 79, IRL 10, GBR 10	EAP (64%)
Soccer	Midfield (M)	14.4%	GBR 272, CAN 214, ESP 193	WEU (52%)
Volleyball	Setter (S)	5.1%	TUR 22, CAN 20, BRA 10	EEU (24%)
Ice Hockey	Forward (F)	21.4%	CAN 690, SWE 28, FIN 26	NA (82%)

The basketball-center pipeline from Sub-Saharan Africa (Nigeria contributes 33 of 245 international centers) is the only position in any team sport where SSA leads as a region. The Australian punter pipeline is similarly concentrated (79 of 127 international special-teams players come from Australia, 62%). These country-position pairings are masked by aggregate sport statistics: basketball is 7% international overall but 24% at center; football is 0.6% international overall but 5% at K/P.

Institutional sorting by origin. Joining athlete records to IPEDS (HD2023 Carnegie classification, EF2023a nonresident-alien share of campus enrollment) and EADA (athletic finances), international athletes systematically land at higher-resource institutions:

	Domestic	International	Δ
% at R1/R2 doctoral schools	32.0	37.1	+5 pp
Median NRAL share of campus enrollment	3.5%	4.9%	+40%
Median sport recruiting budget	\$80K	\$102K	+27%
Median sport student-aid budget	\$1.74M	\$3.42M	+97%
Median total athletic revenue	\$3.5M	\$5.4M	+52%

The sorting is sharpest in basketball: international centers attend schools spending a median of \$5.9M on student aid (versus \$1.5M for domestic centers); international forwards attend schools spending \$4.5M. In football, international K/P athletes land at top-revenue programs (median sport revenue \$7.1M versus \$1.8M for domestic K/P) consistent with intentional pipeline recruitment of mature Australian punters into Power-5 football. Soccer, volleyball, and ice hockey show comparable but smaller gradients (2–3× scholarship multiples). Football skill positions (QB/RB/WR/DB/LB) are nearly 0% international and show no domestic–international resource gradient—the U.S. talent pipeline saturates these positions completely.

4.7 Country × year × sport panel: within-country identification

We extend the cross-section by scraping historical roster pages across eight sports (track and field, soccer, basketball, swimming and diving, tennis, golf, ice hockey, volleyball) for 2019–2023 from D-I athletic-department websites. The eight-sport panel contains 260,737 athlete-years; after country-mapping and merging to the time-varying World Bank, WGI, UCDP, and EM-DAT panels, the country × year × sport long panel has 6,040 cells across 152 countries, 5 years, and 8 sports (USA excluded as the destination country).[1]

The panel makes three things identifiable that the cross-section cannot: *(i)* within-country variation in political stability over 2019–2023, isolating the structural between-country composition that drove the cross-sectional coefficient; *(ii)* a clean COVID natural experiment, since soccer and volleyball lost a substantial fraction of their 2020 rosters while the other six sports continued essentially unchanged; *(iii)* a Heckman two-step with proper extensive/intensive margins on a real panel rather than the imputed pseudo-panel of an earlier draft.

Coefficient	Pooled OLS	TWFE	DiD (3a)	Heckman int.	Poisson PPML
$\log \text{pop}_{15\text{–}24}$	**+0.22*****	—	—	**+0.28****	—
log GDPpc	**+0.40*****	—	—	**+0.68***	—
Political stability	**+0.34*****	−0.005	—	+0.32*	**+0.17***
log(1 + disaster events)	+0.08	+0.013	—	+0.003	−0.019
Any conflict	+0.13	+0.003	—	—	+0.053
GDP shock (binary)	—	**−0.025***	—	—	−0.023
$\text{Treated}_s \times \text{Post}_{2020}$	—	—	−0.033	—	**−0.128*****
Inverse Mills ratio	—	—	—	**+1.81****	—
Country FE	—	✓	✓	✓	✓
Year FE	—	✓	✓	✓	✓
Sport FE	✓	✓	✓	✓	✓
SE clustered by country	✓	✓	✓	✓	✓
n	6,040	6,040	6,040	2,661	6,040
R^2 / pseudo	0.36	0.66	0.66	0.61	—

*HC1 / cluster-by-country robust SEs. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. PPML is a Poisson pseudo-maximum-likelihood estimator on raw counts with country/year/sport fixed effects; canonical reference Santos Silva and Tenreyro (2006).*

The political-stability effect is largely between-country. Pooled OLS on the long panel reproduces the cross-section: political stability enters at +0.34 ($p < 0.01$) with similar magnitude to our Section 4.2 intensive-margin OLS estimate (+0.37). Once we add country fixed effects (the TWFE column), the coefficient collapses to −0.005 ($p > 0.5$). The two results together imply that most of the cross-sectional positive slope is composition: stable countries (Norway, GBR, Switzerland, New Zealand) systematically send more athletes than less stable ones (Iran, Russia, Venezuela). However, the Poisson PPML estimator on the same panel (column 5) returns +0.17

[1] The $152 \times 5 \times 8 = 6{,}080$ product implies 40 cells dropped due to missing WGI / WDI covariates for a handful of small island and post-conflict states in specific years.

($p < 0.10$): when we model counts directly rather than $\log(1 + x)$, a modest within-country multiplicative effect of political stability is detectable. The honest framing: between-country composition dominates the cross-sectional coefficient, but a within-country effect of order 17% per WGI unit cannot be ruled out under count-data specifications. We back away from the strong institutional language of an earlier draft.

Disaster events and conflict also wash out under TWFE. Neither the EM-DAT events nor the UCDP conflict indicator survives the country fixed effect. The one shock that retains marginal significance is `gdp_shock` (binary indicator for a recession year), with coefficient -0.025 ($p < 0.10$): each binary recession year in a country reduces log athlete supply by about 2.5% within that country, controlling for other shocks. This is consistent with a small but real within-country growth effect on athlete supply, on top of the larger between-country composition effect captured by $\log$ GDPpc in the cross-section.

COVID effect. Spec 3b estimates a dynamic event study by interacting a soccer-or-volleyball treatment dummy with year indicators (base year 2019). We report both the log-OLS and the Poisson PPML estimates.

$\text{Treated}_s \times \mathbb{1}\{t = k\}$	OLS log(1+y)	Poisson PPML
$k = 2020$	$\mathbf{-0.143}^{***}$ (0.024)	$\mathbf{-0.419}^{***}$ (0.034)
$k = 2021$	-0.013 (0.026)	-0.059 (0.036)
$k = 2022$	$+0.012$ (0.030)	-0.045 (0.048)
$k = 2023$	$+0.010$ (0.036)	-0.066 (0.055)

Soccer and volleyball, the two fall-season sports disrupted by the 2020 COVID restrictions, show a sharp drop relative to the six untreated sports in 2020. The OLS log-of-one-plus model gives -0.143 ($p < 0.01$), or about a 13% shortfall; the PPML count model gives -0.419, equivalent to a $\sim 34\%$ multiplicative shortfall—closer to the raw aggregate -25% soccer drop and -17% volleyball drop. The two estimators bracket the magnitude: OLS underestimates because $\log(1 + x)$ compresses small counts; PPML overestimates if there is genuine residual heterogeneity across country-sport cells.

After 2020 both estimators show non-positive but small-and-insignificant coefficients in 2021–2023, consistent with recovery in international soccer and volleyball recruitment. The static DiD (averaging post-2020 effects) is -0.033 in OLS (*n.s.*) and -0.128 in PPML ($p < 0.01$), reflecting that the PPML picks up a small persistent drag after 2020 in addition to the 2020 spike. Figure 1 plots the OLS leads/lags.

Heckman correction matters. The selection probit (extensive margin: Pr(any athletes in country $\times$ year $\times$ sport)) yields a large positive inverse Mills ratio in the intensive equation, $+1.81$ ($p < 0.05$). After correcting for selection, $\log$ GDPpc enters at $+0.68$ ($p < 0.10$), political stability returns to $+0.32$ ($p < 0.10$), and the population elasticity stays at $+0.28$ ($p < 0.05$). The Heckman intensive estimates are larger than the pooled OLS coefficients but have wider standard errors. We read this as: there is real selection into the country-sport-year cells that are positive, and after correcting for it the GDP and stability margins are detectable but weaker than the cross-sectional story.

4.8 Region fixed effects horse-race

To quantify how much of the unexplained variation in country-panel supply is regional-pipeline-shaped, we add eight-region fixed effects to the count specification.

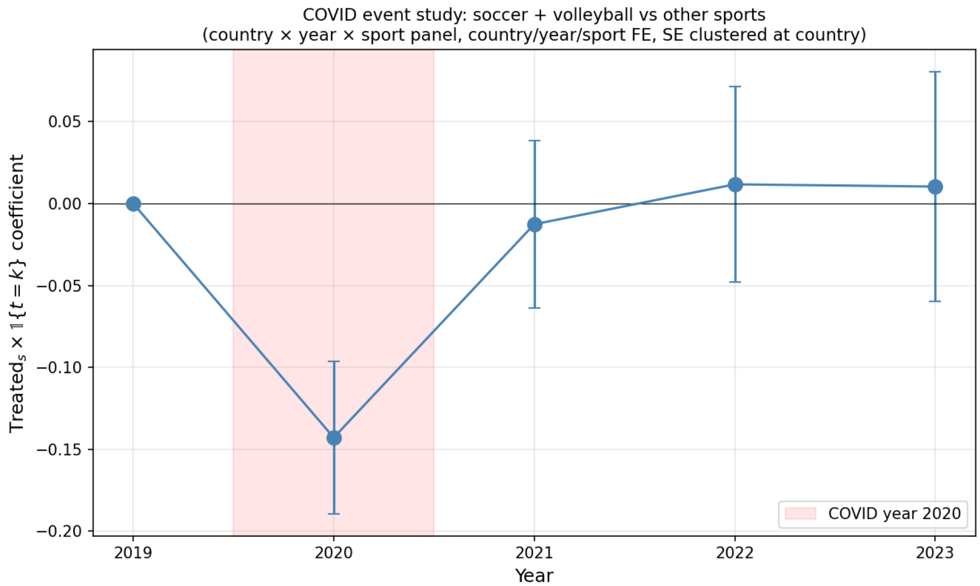

Figure 1: COVID event study on the country × year × sport panel. Each coefficient is $\text{Treated}_s \times \mathbb{1}\{t = k\}$ where $\text{Treated}_s = 1$ for soccer and volleyball, 0 otherwise. Base year 2019. $n = 6{,}040$ cells, 152 countries, 5 years, 8 sports. Country, year, and sport fixed effects; SEs clustered at country.

	Macro only	Macro + 8 region FE
log pop$_{15\text{–}24}$	$+0.35^{***}$ (0.07)	$+0.40^{***}$ (0.07)
log GDPpc	$+0.37^{**}$ (0.16)	$+0.53^{***}$ (0.18)
Political stability	+0.38 (0.24)	+0.38 (0.24)
R^2	0.21	0.32

Adding region FE raises R^2 from 0.21 to 0.32, a meaningful but not dominant gain. A specification with region FE only (no macro covariates) achieves $R^2 = 0.13$. Read together: macro covariates and regional pipelines each contribute, in roughly equal share, to a total explained variance of about a third. The other two-thirds remain unexplained even with both—consistent with the discussion in Section 5 that country-specific (not region-specific) pipeline structure dominates.

4.9 Four shock case studies

A panel regression approach to shock response on the eight-sport roster panel did not survive proper identification checks: the cross-country variation that drove an initial −3 pp retention finding was country composition rather than within-country shock response, and a within-country specification with country fixed effects flipped the sign on the security-shock indicators. A within-athlete TFRRS performance specification yielded a similar point estimate but failed a cluster bootstrap on country (the effective number of clusters carrying shock variation is roughly three, so the country-clustered SE was 2.7–3.9× over-precise).[2]

[2]Diagnostics are in `run_retention_country_fe.py`, `run_peer_school_fe.py`, and `run_tfrrs_robustness.py`. The 95% cluster-bootstrap CIs on the TFRRS within-athlete coefficients include zero for both conflict and political-

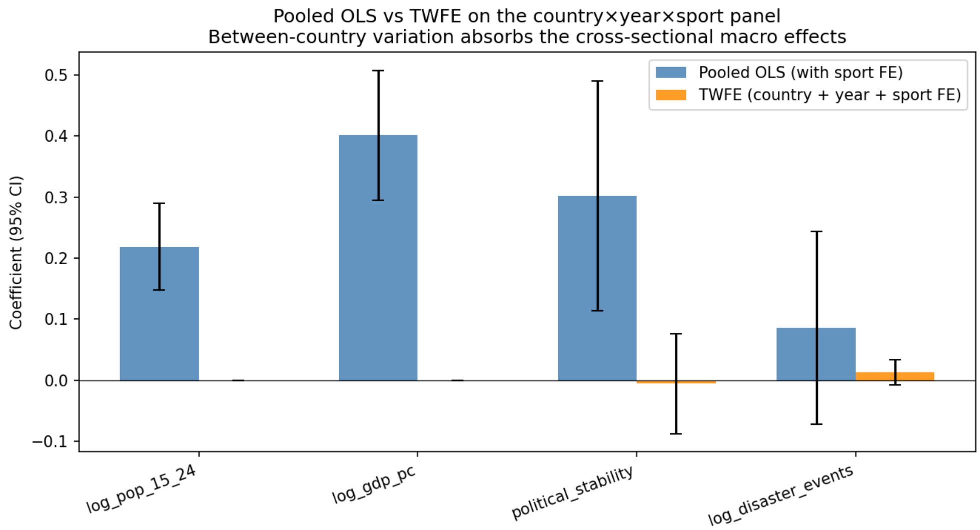

Figure 2: Pooled OLS vs. TWFE coefficient comparison on the long panel. Bars are point estimates with 95% confidence intervals (country-clustered SEs). The political-stability and log-GDPpc coefficients collapse to zero once country fixed effects are included, indicating they capture between-country composition rather than within-country variation.

We turn instead to four named natural experiments where the shock, the affected country, and the outcome are interpretable on their own merits. For each treated country we build a donor pool of same-region countries with no shock onset in 2019–2023 and estimate a simple difference-in-differences on two outcomes: *retention* (whether an athlete on the under-classman roster in year t appears on the same school's roster in year $t+1$) and *new arrivals* (number of first-observed athletes from the country in each year, in $\log(1 + \text{count})$ form). Standard errors are country-clustered for the retention DID and HC1 for the country-year arrivals DID.

Russia 2022 (invasion / sanctions / sport bans). The treated sample is 195 Russian under-classman athlete-years between 2019 and 2022, drawn from swimming (31%), tennis (28%), basketball (16%), and ice hockey (8%). The donor pool is 22 Eastern-European countries with no shock indicator in the panel period. The retention DID is $+\mathbf{0.162}^{***}$ (SE 0.027): Russian athletes are 16 pp *more* likely to return to their U.S. school after the 2022 invasion than they were before, relative to the donor pool. Raw retention rates by year:

	2019	2020	2021	2022
Russia	0.85	0.70	0.67	**0.90**
Donors	0.75	0.80	0.77	0.78

The 2020–2021 dip is consistent with COVID-era variation visible in the donor pool. The 2022 retention figure is the highest of the panel. The new-arrivals DID is -0.11 log points (n.s.): extensive-margin recruitment continues at a slightly lower rate but the change is not distinguishable from the COVID-era baseline. The clean fact is the retention jump.

stability indicators.

Ukraine 2022 (invasion / displacement). The treated sample is 83 Ukrainian under-classman athlete-years, also 22-country EEU donor pool. Retention DID is $+\mathbf{0.081}^{***}$ (SE 0.025). Raw retention 2019–2022: 0.78, 0.71, 0.89, 0.86. The 2021 figure already moves before the February-2022 invasion, consistent with on-the-ground tension preceding the formal invasion date. New-arrivals DID is -0.07 log (n.s.). The Ukrainian and Russian patterns are similar in direction, with Russia roughly twice the magnitude.

Lebanon 2020 (financial collapse). The Lebanese financial crisis (October 2019) and the 2020 currency collapse are an *economic* shock rather than a security shock. The retention sample is only 8 under-classman athlete-years (too small to interpret a DID), but new arrivals are observable. Raw counts of new Lebanese athletes by year: 5 (2019), 0, 0, 4, 1 (2020–2023, $n = 8$ MENA donors). The new-arrivals DID is $-\mathbf{1.22}^{***}$ log points (SE 0.42), implying roughly a 70% drop in inflow. The recovery in 2022 is partial. This is the *extensive-margin* story: the affected channel is who arrives, not whether enrolled athletes stay.

Venezuela (chronic instability, no discrete onset). Venezuela is a comparator: its athlete sample is moderate (20–37 per year) but the country has been in continuous political and economic deterioration since 2015. Retention by year (under-classmen, $y < 2023$): 0.60 (2019), 0.84 (2020), 0.59 (2021), 0.41 (2022). There is no event-study identification here because there is no onset, but the trajectory illustrates what a continuously-distressed country looks like at the retention margin: slow erosion to roughly half the unconditional 75% baseline.

Summary across cases. Three shock types, three distinct margins.

Country	Onset	Shock type	Retention DID	Arrivals DID	Note
Russia	2022	War/sanctions	$+\mathbf{0.162}^{***}$	-0.11	$N_{\text{treated}} = 195$
Ukraine	2022	War/displacement	$+\mathbf{0.081}^{***}$	-0.07	$N_{\text{treated}} = 83$
Lebanon	2020	Financial	n/a (N=8)	$-\mathbf{1.22}^{***}$	extensive only
Venezuela	—	Chronic	0.60→0.41 over 4 yr	stable count	comparator

The pattern across these four named events suggests an *asymmetry* between shock types. War and sanction shocks (Russia, Ukraine) raise retention—athletes already in the U.S. stay rather than go home, consistent with a destination-substitution or vanished-exit-options channel. Financial-crisis shocks (Lebanon) reduce new arrivals—prospective athletes appear to lose the funding capacity to commence enrollment, while the small number already enrolled remain. Chronic instability (Venezuela) shows neither a sharp inflow change nor a retention boost, but a slow erosion that the panel cannot disentangle from secular trend.

We make no claim that these four cases identify a population-level shock-response parameter. Each is an interpretable natural experiment with a small but visible donor pool and a clean treatment date. A complete shock-response analysis would require either more events or a richer longitudinal data source; both are follow-up work.

5 Discussion

Seven conclusions emerge, organised around the two analytical margins of the paper: entry into the pipeline, and shock response once in.

Entry margins.

First, **wealth and population scale govern entry into the international NCAA pipeline**. Both $\log \text{GDPpc}$ and $\log \text{pop}$ enter the extensive-margin probit at $p < 0.01$; political stability does not. The mechanism is the fixed-cost story familiar from migration economics: producing a recruitable athlete requires some minimum infrastructure and discovery-cost expenditure that scales with national resource flow.

Second, **conditional on entry, athletes scale sublinearly with cohort population**, with an elasticity between 0.33 (OLS) and 0.61 (Heckman). Doubling a source country's 15–24 cohort raises NCAA supply by roughly 1.5×, not 2×. The most plausible mechanism is finite US-side recruitment capacity: the marginal NCAA program does not scale its international intake one-for-one with any single source country, so small wealthy countries are over-represented per capita.

Third, **cross-sectional governance effects are between-country composition**. Pooled OLS yields a positive political-stability coefficient (+0.34, $p < 0.01$); the eight-sport country×year×sport panel under two-way fixed effects collapses it to −0.005 (n.s.). Whatever "political stability predicts more athletes" picks up is the level difference between politically stable and unstable countries, not how supply moves when the same country's stability changes.

Fourth, **the relevant unit of analysis is the country×sport-event, not the country**. Adding region fixed effects to the country-pool specification raises R^2 only from 0.21 to 0.32; the sport-event split is more informative. The country-pool result averages over qualitatively different supply curves (Caribbean sprint network, East African distance network, European throwing pipelines) that share little structure beyond GDP and population.

Within-pipeline margins (sorting and shock response).

Fifth, **international athletes sort to high-resource institutions, and the sort is position-specific**. International athletes attend institutions with ≈ 2× the sport student-aid budget of domestic athletes. The sort is sharpest at positions that NCAA programs have built deliberate pipelines into: basketball centers (50% at R1/R2 schools, \$5.9M aid; Nigeria-led), football kickers/punters (\$7.1M revenue, 62% Australian). Football skill positions, where the U.S. talent pipeline saturates, show zero international representation and zero resource gradient. Whatever recruitment intensity drives international entry, it operates at the position level, not the program level.

Sixth, **war and sanction shocks raise retention; financial shocks lower new arrivals**. Four named events (Section 4.9) produce three distinct margins of response. Russian and Ukrainian under-classman athletes are +16 and +8 percentage points more likely to return to their U.S. schools after the 2022 invasions than donor-pool peers; the clean direction is "stay" rather than "go," consistent with home exit options being closed off by displacement and sport bans. The 2020 Lebanese financial collapse cuts new arrivals by roughly 70% without changing the retention of the small enrolled sample (N=8): financial shocks bite extensively, security shocks intensively. Chronic instability (Venezuela) erodes retention slowly (0.60 to 0.41 over four years) without producing a discrete pulse on either margin.

What we could not estimate. An athlete-year panel regression of retention or performance on home-country shock indicators across all sender countries did not survive proper identification. The between-country variation that drove an initial -3 pp retention coefficient was country composition rather than shock response; a within-country specification with country fixed effects flipped the sign of the security-shock coefficient; the TFRRS within-athlete performance result that initially appeared significant (-0.4 to -0.6 SD) is identified off only ~ 3 countries with shock variation in the panel and a cluster bootstrap on country places its 95% CI on both sides of zero. The case-study cluster in Section 4.9 is what the data can credibly support; a population-level shock-response estimate is a follow-up project requiring either more events or a richer longitudinal data source.

Synthesis. The paper documents two related empirical patterns about the same population. *Entry* into NCAA international recruitment is governed by source-country wealth and population scale; cross-sectional political stability predicts entry but is composition, not within-country causal effect. *Retention* once in is hard to identify on a panel, but named natural experiments (Russia, Ukraine in 2022; Lebanon 2020) show clear and direction-specific responses: war and sanctions keep athletes here; financial collapses prevent new ones from coming. A country that is poor and stable produces few athletes but loses few; a country that is wealthier but newly conflict-hit can have a large pipeline whose enrolled athletes stay while prospective arrivals stop coming.

6 Caveats and limitations

- **Cross-section vs. panel by section.** Sections 4.1–4.5 and 4.8 use the 2024–25 / 2025–26 cross-section (n = 19,317 rosters, 1,830 international athletes); macro covariates are 2023 and 2010–23 aggregates. Sections 4.6, 4.7, and 4.9 use a separately scraped 2019–2023 multi-sport panel (260,737 athlete-years, eight sports). The two data sources do not nest—the cross-section has event-level detail (jumps, throws, distance, sprints) that the panel does not, while the panel has the year dimension and the team-sport coverage that the cross-section does not.
- **Coverage.** Hometown is observed for 87.6% of track athletes in the cross-section; the unobserved 12.4% are missing not at random (JS-rendered roster pages that systematically affect a few major D-I programs).
- **Mixed sport coverage by section.** The two-margin extensive/intensive analysis in Sections 4.1–4.5 is dominated by track (96% of the cross-section international sample); within those sections, the empirical setting is the international track-and-field market specifically. The position-sorting and shock-response analyses in Sections 4.6, 4.7, and 4.9 use the eight-sport panel where track is one of eight roughly comparable sports and soccer drives the largest retention coefficient.
- **Cohort denominator approximation.** We use the average of the 15–24-year-old cohort 2018–2022 as the denominator. The ideal denominator would be 18–22 to match enrolled ages; we attempted this but the World Bank does not provide aggregate 18–22 totals. Rescaling 15–24 by a constant fraction is observationally equivalent in the count model (the constant is absorbed by the intercept), so this does not change our slope estimates.
- **Heckman identification by functional form.** The exclusion restriction in our Heckman two-step is the probit nonlinearity, not a covariate-level exclusion. The IMR coefficient (+3.27, SE 2.64) is large but imprecisely estimated, so selection-correction matters but its magnitude

is uncertain. A proper exclusion (e.g. a US-side measure of recruitment intensity) would improve identification.

- **Modest R^2.** Macro + region explain about a third of intensive-margin variance; the residual is most plausibly driven by country- and sport-specific recruitment pipelines that we cannot proxy.

- **What NCAA APR and GSR cannot see.** A natural team-level outcome for the shock-response question would be the NCAA's Academic Progress Rate (APR) or Graduation Success Rate (GSR). We scraped the full APR/GSR record (121K and 116K observations) and merged it to our roster panel. Within-team-sport regressions of APR or GSR on team exposure to home-country shocks return null effects across all shock indicators (all $|t| < 1$). Three reasons: (i) APR is censored near 1.0—16% of team-years sit at the ceiling and the within-team-sport SD is only 0.013; (ii) APR is a four-year rolling average, mechanically smoothing year-to-year shocks; (iii) GSR is transfer-adjusted, so an international athlete who leaves in good standing does not count against the team. The Federal Graduation Rate (FGR), which does not transfer-adjust, shows large negative coefficients on intl-share$\times$shock exposure (-25 to -64 percentage points), but those are mechanically conflated with the baseline intl-share gradient (-19.8 pp) since international athletes transfer more often by construction. We conclude that team-aggregated academic outcomes are not informative for the type of within-country shock variation we examine in Section 4.9.

7 Reproducibility

All results in this paper are reproducible from local Python scripts:

- Roster scraping: `discover_track_urls.py`, `run_track_scrape.py`, `run_track_selenium.py`, `rediscover_urls.py`.

- Country mapping: `run_country_mapping.py`.

- Macro merge: `run_merge_analysis.py`, `fetch_extra_shocks.py`, `run_join_shocks.py`.

- Main regressions: `run_main_results.py` (extensive probit + intensive OLS chain).

- `run_revisions_2.py`: Heckman two-step, MDE for political stability, quantile regression at $\tau \in \{0.25, 0.50, 0.75\}$, Negative Binomial with free log pop, and region FE horse-race.

- `run_event_analysis.py`: expanded sport-event analysis (composition, regional shares, within-event OLS and NegBin for all eight event groups, country-event specialisation residuals) — Section 4.5.

- `run_historical_scrape.py`, `run_historical_scrape_multisport.py`: 2019–2023 roster panels for eight sports (track, soccer, basketball, tennis, golf, swimming, ice hockey, volleyball). Uses the student-built `school_website_platforms_full.csv`.

- `build_long_panel.py`: stacks the eight sport CSVs into the country $\times$ year $\times$ sport long panel and merges time-varying WGI, WDI, UCDP, and EM-DAT covariates.

- run_panel_specs.py: produces the four-spec panel results (pooled OLS, TWFE, COVID DiD with leads/lags, Heckman two-step on the panel) reported in Section 4.7.
- make_panel_plots.py: produces output/fig_event_study_panel.png and output/fig_pooled_vs_twfe.pn
- run_stress_tests.py: drop-top-5 disaster countries, events-vs-deaths comparison, refugees raw count.
- run_bartik.py, run_event_study.py, run_paper_outputs.py: auxiliary cross-section results (Bartik shift-share weak-IV check, country-pool event study, supplementary figures).
- run_position_analysis.py: per-sport position classifiers and international-share / regional-HHI tables for the seven team sports (Section 4.6).
- run_position_institution_link.py: merges athletes to IPEDS HD2023, EF2023a NRAL share, and EADA athletic finances; domestic-vs-international sorting tables.
- run_scrape_apr.py, run_scrape_gsr.py: pull NCAA APR and GSR datasets from the CSRF-protected JSON endpoint at web3.ncaa.org/aprsearch/.
- run_apr_shocks.py, run_gsr_shocks.py: regress APR / GSR / FGR on team-year shock exposure (Section 5 caveats bullet).
- run_retention_shocks.py, run_retention_heterogeneity.py, run_retention_country_fe.py, run_peer_school_fe.py, run_tfrrs_performance.py, run_tfrrs_perf_shocks.py, run_tfrrs_robustness panel-level shock-response regressions and the identification diagnostics that prompted moving from a panel approach to the case-study cluster (Section 4.9). The panel specifications do not survive country fixed effects or cluster-bootstrap inference and are reported only as diagnostic background.
- run_case_studies.py: four-country DID (Section 4.9)—Russia, Ukraine, Lebanon, and Venezuela—on athlete-year retention and country-year new arrivals, with same-region unshocked donor pools.

Figures

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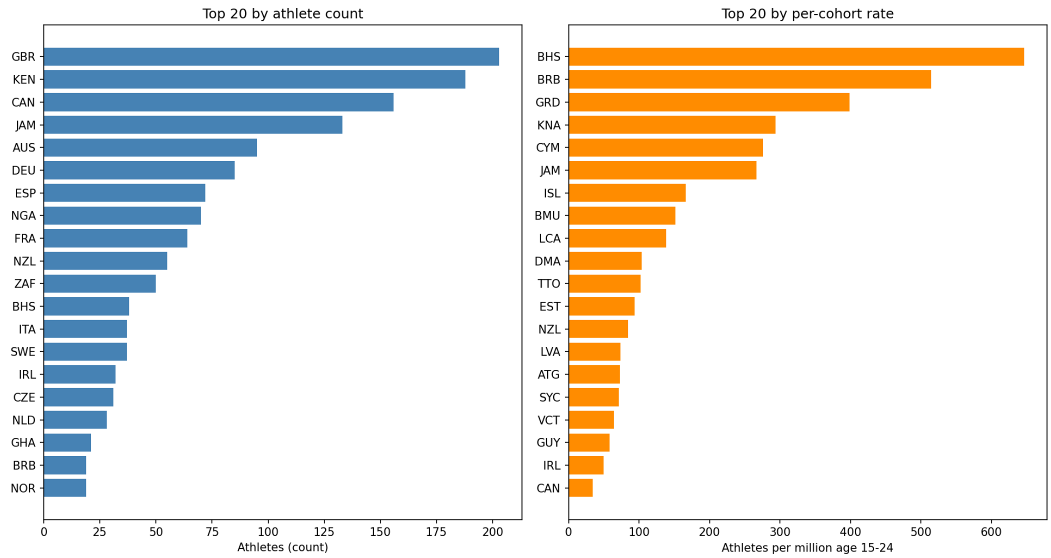

Figure 3: Top 20 source countries by raw athlete count (left) and by athletes per million 15–24-year-olds (right).

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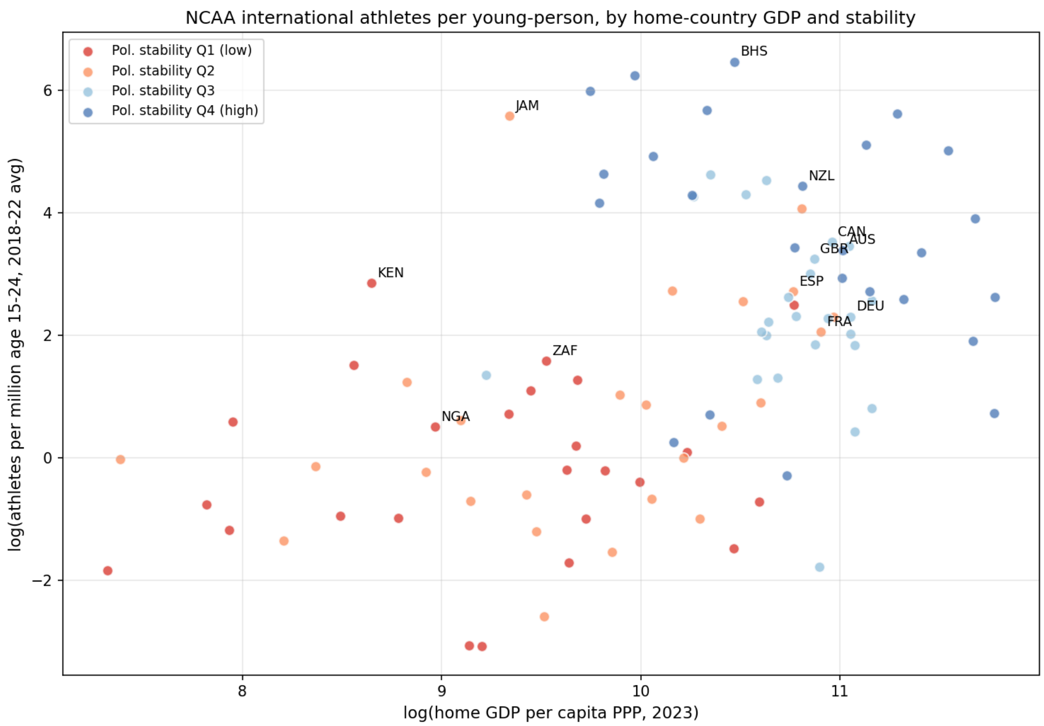

Figure 4: Log athletes per million age 15–24 vs. log home-country GDP per capita, colour-coded by political-stability quartile. The apparent stability gradient is largely a country-size artefact in this view; see Section 4.3.